BMIT3173 Integrative Programming

ASSIGNMENT SPECIFICATION

Introduction

Students will work in a team comprising 4 to 5 members from the same practical class as this will provide an appreciation of the effects of scale that is present when creating a non-trivial system and how various technologies may be integrated to create a complete system. Each student will be individually assessed based on the quality and scope of work completed by him/her.

Learning Outcomes Assessed

CLO	Description	Contribution to Coursework
CLO2	Construct programs that use integrative technologies (P4, PLO3).	85%
CLO3	Demonstrate self-advancement through independent learning of integrative programming technologies (A3, PLO9).	15%

Problem Statement

Objective

Students are required to select a real or hypothetical organisation whose mission, products, or services align with **one (1) of the United Nations Sustainable Development Goals (SDGs)** as the focus of their project. The chosen organisation has tasked your team with developing an innovative web-based solution that contributes to the selected SDG while optimizing business operations, enhancing user engagement, improving service delivery, and driving overall performance.

Each team must choose **one (1) SDG** from the list below as the thematic focus of the project. The selected organisation (real or hypothetical) must have a mission, product, or service that meaningfully contributes to the chosen SDG. Refer to the official UN SDG portal at https://sdgs.un.org/goals for full descriptions of each goal and its targets.

SDG 1: No Poverty
SDG 2: Zero Hunger
SDG 3: Good Health and Well-being
SDG 4: Quality Education
SDG 5: Gender Equality
SDG 6: Clean Water and Sanitation
SDG 7: Affordable and Clean Energy
SDG 8: Decent Work and Economic Growth
SDG 9: Industry, Innovation and Infrastructure
SDG 10: Reduced Inequalities
SDG 11: Sustainable Cities and Communities
SDG 12: Responsible Consumption and Production
SDG 13: Climate Action
SDG 14: Life Below Water
SDG 15: Life on Land
SDG 16: Peace, Justice and Strong Institutions
SDG 17: Partnerships for the Goals

Examples (non-exhaustive): an NGO promoting clean water access (SDG 6); an online learning platform for underserved communities (SDG 4); a renewable energy services startup (SDG 7); a tele-health or mental health support service (SDG 3); a platform supporting recycling and the circular economy (SDG 12); a microfinance platform for low-income communities (SDG 1); a women's career and mentorship network (SDG 5); a smart-city waste management system (SDG 11). Each team should select a topic whose context allows meaningful integration of all the technical requirements stated in the Scope of Work below.

Scope of Work

Initial Deliverables (Due Week 4 - Group Submission)

1. **System Description**
   Provide a brief description of the system that you will be developing.
   State the chosen SDG and provide a brief justification (2–3 sentences) explaining how the proposed system contributes to that SDG.

2. **Analysis Class Diagram**
   - Submit an analysis class diagram (including entity classes with relationships, excluding methods, attributes, and interfaces).

3. **Team Member Modules**
   - Each team member should outline the module(s) they are responsible for, explaining the functions and scope of their module(s).

Individual Responsibilities (Due to week 12):

Each student must:

1. Implement at least **one (1) module** of the system.

2. Ensure that your module design and implementation incorporate the following technologies and practices:
   - **PHP and MySQL**
   - **Design Patterns**
   - **Secure Coding Practices**
   - **Web Service Technologies**

3. The **entire system** must follow the **Model-View-Controller (MVC) architecture**, and you are required to use **Object Relational Mapping (ORM)** in your program.
   Note: Marks will be deducted if the above requirements are not strictly followed.

4. Using a framework (e.g., Laravel, CodeIgniter) is optional, but students are **encouraged** to utilize a framework to improve the organization, scalability, and maintainability of their code.
   Note: Frameworks provide built-in tools and libraries that can help streamline development, enhance security, and speed up the process of building robust, feature-rich applications

Please refer to the **Assignment Rubrics** for detailed assessment criteria and the allocation of marks.

Submission Deadline

- Sunday, 6th September 2026 (Week 12)
  - Assignment report
    - Each student must prepare an individual assignment report using the **provided document report template**.
    - The completed **AI Usage Disclosure Form** (included in the report template) must be completed and submitted as part of the report, whether AI tools were used.
    - Include the following section of the assignment:
      - **PHP and MySQL**
      - **Design Patterns**
      - **Secure Coding Practices**
      - **Web Service Technologies**
    - Refer to the template for detailed instructions on each section.
      - **Submission Format**
        - Convert the report to a **PDF file**, rename it with your **full name**, and submit it to the **Google Classroom assignment.**
  - Team source code – to be uploaded to a specified Google Drive folder. Include:
    - The complete project for your application (group leader to submit).
    - Each file should have the author's name included in the header comment.
    - The complete SQL scripts for creating the database and tables, as well as populating the database tables with initial data.
- You will also be required to give a demo of your completed work to your tutor during Week 13 and 14.

Academic Integrity and Plagiarism

There must be originality in your work, i.e., do not copy or refer to other teams. You may only work with your team members(s) to solve this assignment. You must not share with or refer to any part of the assignment (including the code) of anyone else except your team member(s) and your tutor.

Before submitting your assignment, ensure that you have complied with TAR UMT's plagiarism policy. Any attempts to cheat, plagiarism, collusion, and any other attempts to gain an unfair advantage in the assessment will cause the students concerned to be penalized.

Students found to be dishonest are liable to disciplinary action.

Note on AI Usage:

This assignment is governed by the TAR UMT Policy for the Use of Artificial Intelligence (AI) (PO/111:26) and is classified as **YELLOW** (Limited AI) under the Red–Yellow–Green Framework. AI tools may be used only for clearly defined support purposes such as language refinement of your own text, clarifying taught concepts, debugging assistance on code you wrote, and early brainstorming. They must not be used to generate the PHP/MySQL code, design-pattern implementation, class diagrams, web-service code, or the core analysis being assessed, nor to upload confidential or others' personal data to public AI tools. You remain responsible for the accuracy and originality of your work, and the tutor may ask you to explain or demonstrate any part to verify authorship.

AI Usage Disclosure Form (compulsory): Every student must complete and submit the AI Usage Disclosure Form provided in the report template, whether AI tools were used, by the assignment submission deadline of **Sunday, 6th September 2026 (Week 12)**. A report submitted without a complete Form is treated as an incomplete submission and may not be marked. Where AI tools were used, each use must be listed in the Form and the tool(s) cited in your References in APA (7th edition) format.

Late Submission

Refer to TAR UMT's Guideline on Late Submission of Coursework.